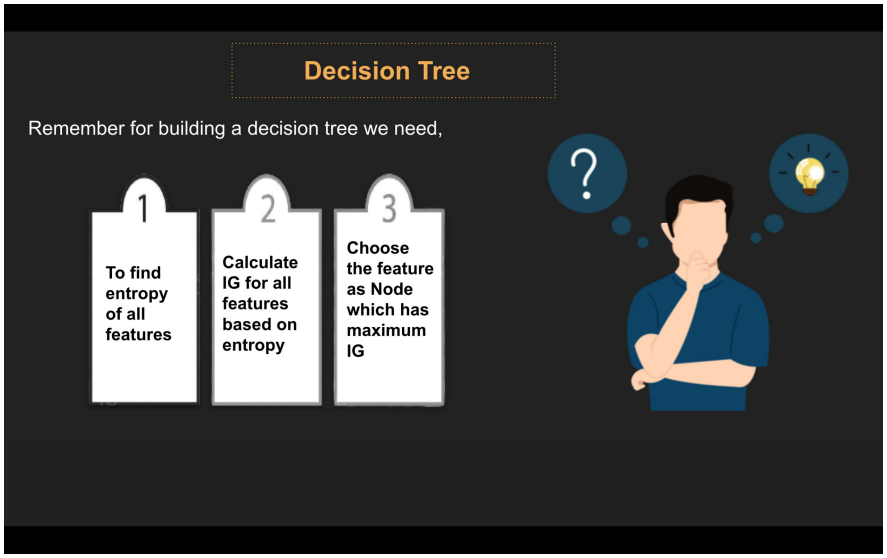
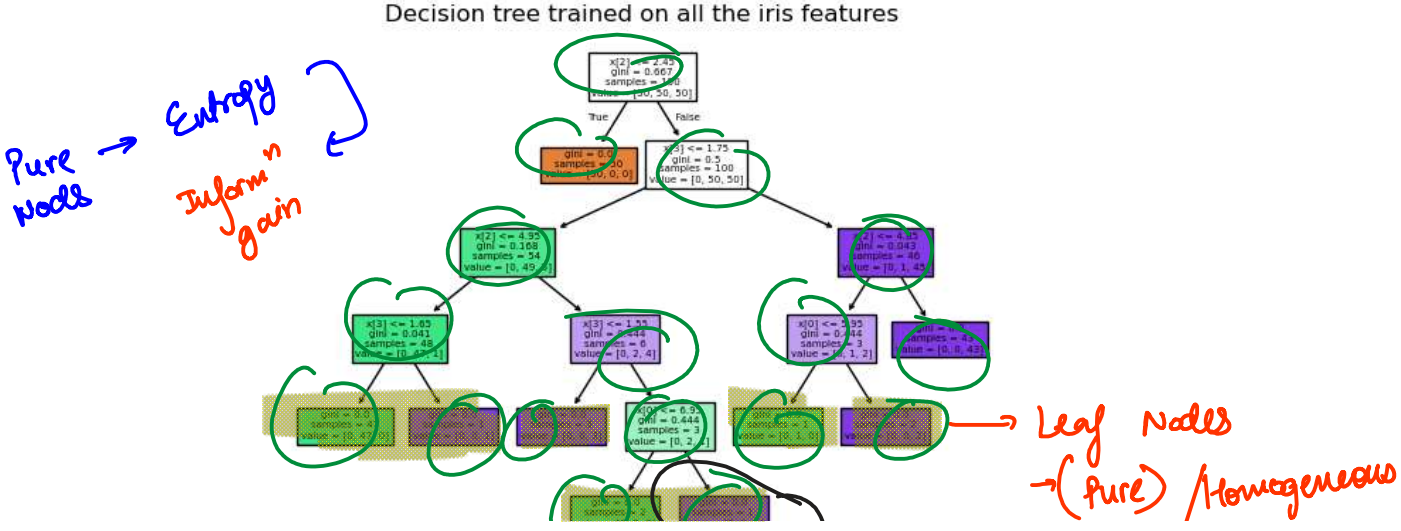
D Trees 2

13 October 2025 21:03



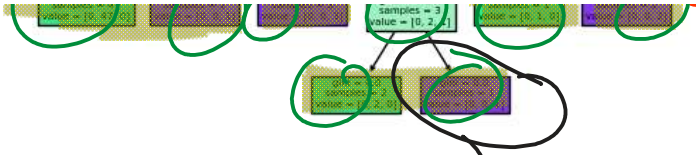
Agenda

- Recap (D. Trees Algo) → Jio employee attrition
- Issue with entropy / Gini Impurity → Log Reg, KNN, → D. Trees
- Minor Details (implementation)
- Feature importance



Pure nodes → Entropy, Inform gain

→ Leaf Nodes → (Pure) Homogeneous



→ (pure) / Homogeneous

entropy (node) → how impure this is

. What Are We Trying to Achieve?

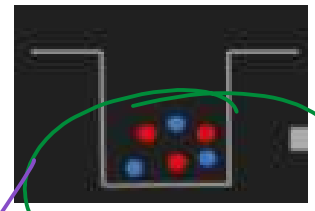
We want **pure (homogeneous) nodes**, i.e. all samples in a node belong to the same class.

At prediction:

- Each leaf node gives a **majority vote**.
- The higher the node's purity → higher our confidence.

Purity → Entropy

(English)



3 Blue

3 Red

How impure this node is ?

$$H = -\sum_{i=1}^{K} p_i \log p_i$$

p_i → prob of ith class

$p_{Red} = \frac{3}{6}$ $p_{Blue} = \frac{3}{6}$

I 3B 3R

II 4B 2R

$$H = -\left[p_{Red}^{1/2} \log_2 p_{Red} + p_{Blue}^{1/2} \log_2 p_{Blue}\right]$$

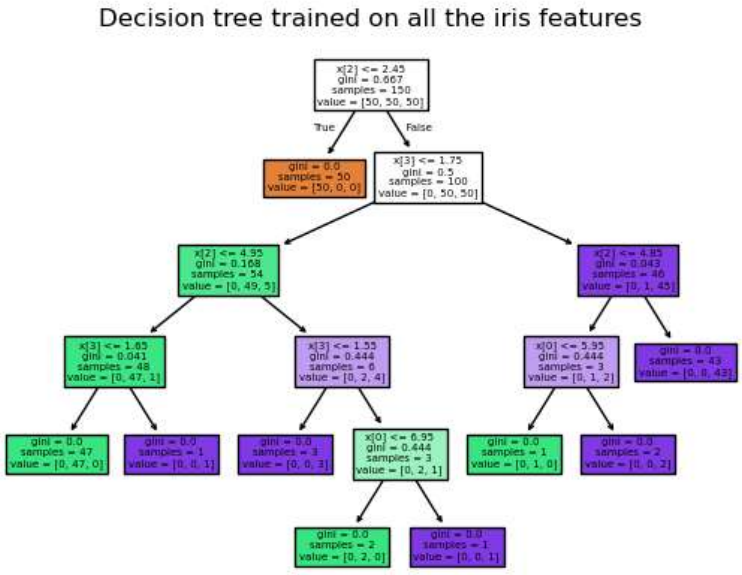
$$\Rightarrow -\left[\frac{1}{2}\log\left(\frac{1}{2}\right) + \frac{1}{2}\log\left(\frac{1}{2}\right)\right] = 1$$

1 → max entropy → impurity

[Algo → Recursive split]

→ select Best feature & threshold which give highest I.G

→ continue till a stopping condition is met



(1) Entropy

(2) Info Gain

Issue with Entropy

$$\{ H(\text{node}) = -\sum p_i \log p_i \}$$

↑ Problematic (computationally expensive)

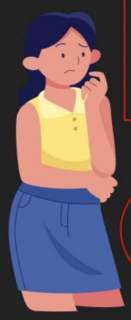
(Gini - Impurity)

Issue with Entropy

Entropy is calculated as: $H(y) = -\sum_{i=1}^{k} p(y_i)log(p(y_i))$

- Notice that we take log of probability.
- We have to do this for each feature, at each node.

This is **computationally expensive** and **time consuming**.

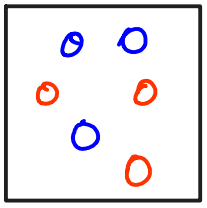


So, what's the problem ?

PROBLEM

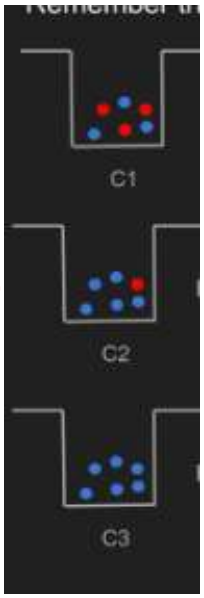
SOLUTION

Gini Impurity



→ Entropy ? = 1 (max)

$$GI = 1 - \sum_{i=1}^{K} (p_i^2) \Rightarrow \left[1 - \left[p_{Red}^2 + p_{Blue}^2\right]\right]$$



	GI	H
C1	$GI = \left[1 - \{(\frac{1}{2})^2 + (\frac{1}{2})^2\right] = 0.5$	1
C2	$GI = 1 - \left[(\frac{5}{6})^2 + (\frac{1}{6})^2\right] = 0.27$	0.65
C3	$GI = 1 - [0^2 + (1)^2] = 0$	0

(IV)

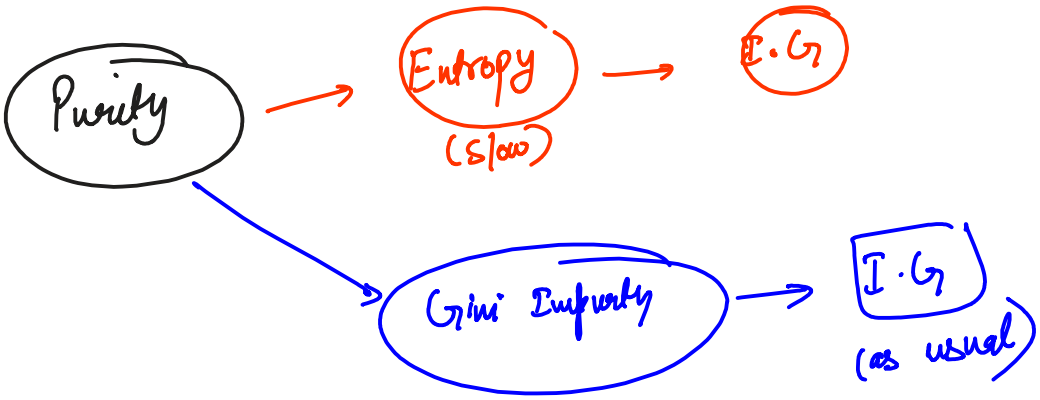
Can we Plot GI → $p_{class\,1} = x$

Can we Plot GI (Binary)

→ P class 1 = x

→ P class 2 = (1−x)

$$GI = 1 - \sum p_i^2 = 1 - [x^2 + (1-x)^2]$$

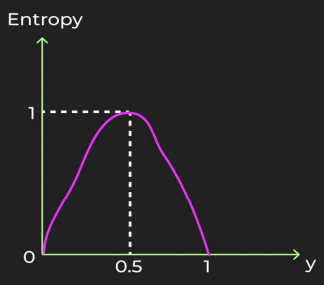


Gini Impurity follows the same trend as entropy :

Meaning → Impurity ⬆ → Entropy ⬆

Difference between Entropy and Gini Impurity :

Range of Entropy [0 , 1]



Range of Gini Impurity [0 , 0.5]



Comparing Gini Impurity and Entropy

Remember the 6 Red and Blue ball example,



C1

$GI(y) = 1 - [(½)^2 + (½)^2] = 0.5$

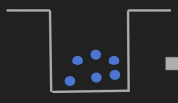
$H(Y) = - [½ \log_2 ½ + ½ \log_2 ½] = 1$



C2

$GI(y) = 1 - [(⅙)^2 + (⅚)^2] = 0.27$

$H(Y) = - [⅚ \log_2 ⅚ + ⅙ \log_2 ⅙] = 0.65$



C3

$H(Y) = 1 - [(0/6)^2 + (6/6)^2] = 0$

$H(Y) = - [1 \log_2 1 + 0 \log_2 0] = 0$

POINTS TO REMEMBER

- Entropy is computationally expensive hence using Gini impurity instead.
- Range of Entropy is [0 , 1] while Gini Impurity ranges [0 , 0.5]



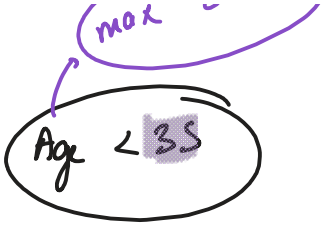
How to split on features

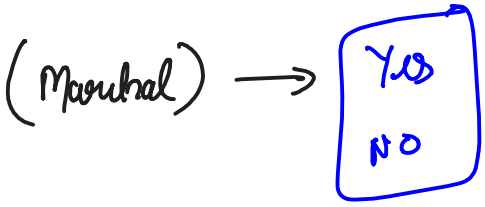
(1) Categorical Feature vs (2) [Numerical features]

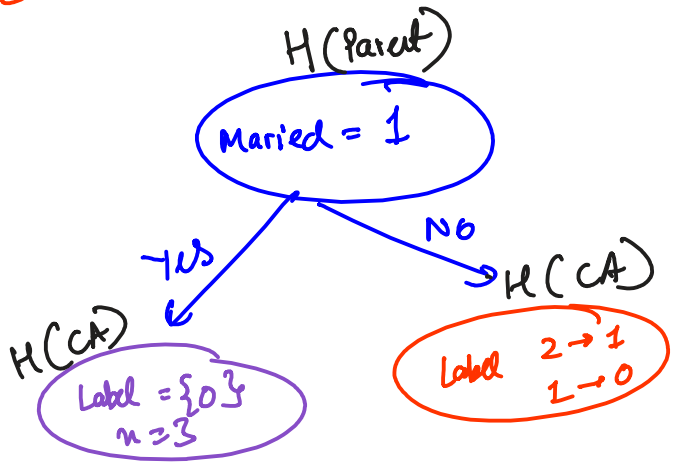
Age	Marital Status	Label (0/1)

max I.G

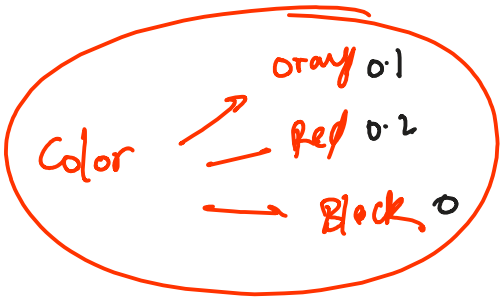
Age	Marital Status	Label (0/1)
22	Yes	0
25	Yes	0
29	No	1
30	No	0
35	No	1
40	Yes	0

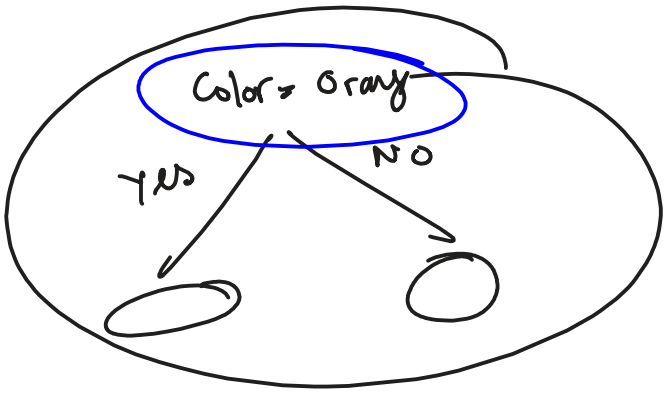


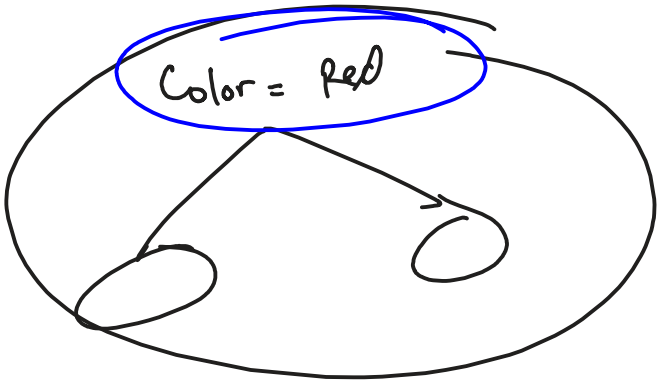


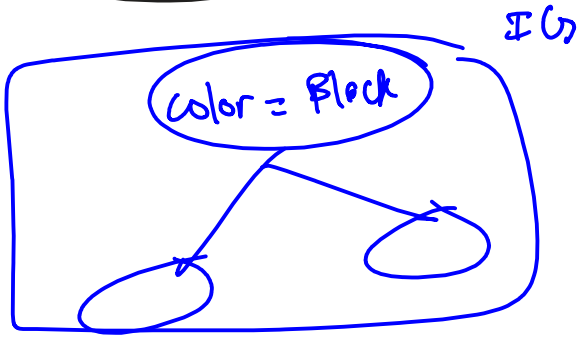


$$IG = H(P) - \left[\frac{n_1}{n} H_A + \frac{n_2}{n} H_B \right]$$





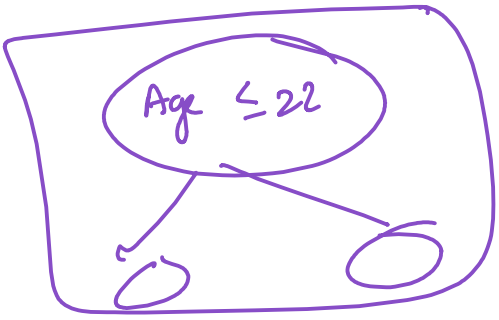


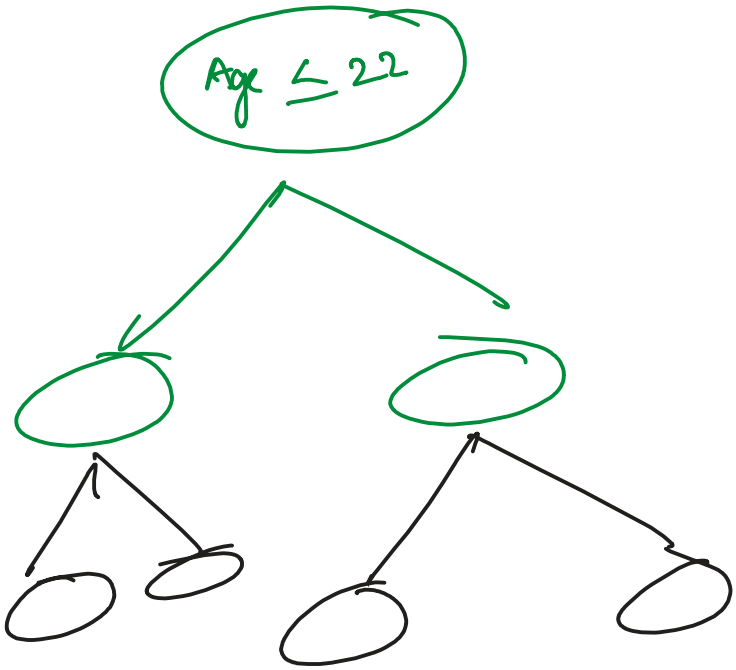


① Arrange Numerical feature in ascending order

Age	IG
22	0.1
23	0.2
25	0
30	0.05
31	0.002

Check all the unique values





What happens when we have a Numerical feature?

DT uses a **threshold-based approach**. Suppose there are 'n' unique values for f_2 features :

	f_2
Unique values	n_1 n_2 . n

why ?

1 Arranging in ascending order

f_2	n_1 →	IG_1
f_2	n_2 →	IG_2
	.	
	.	
f_2	n →	IG_n

2 Take each value (n_1 , n_2 .....n) as threshold and find IG values

$i = \text{argmax}(IG_1, \ldots\ldots IG_n)$
$\text{threshold} = n_i$

3 Choose that threshold which yields highest IG value

**NOTE : If IG is same for future values then randomly choose any

Problem with threshold approach

If there are three numerical features each with 1000 unique values :



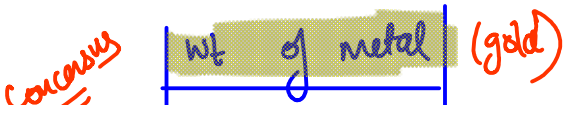
Time complexity = 0 (number of features * unique value)

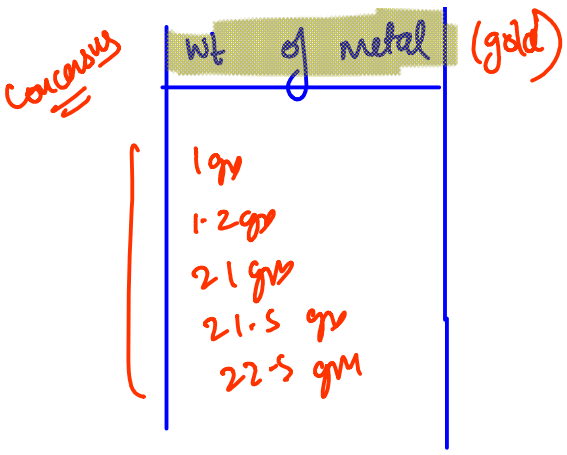
→ Outcome of this will be huge

$$IG = GI_{(parent)} - \left[\frac{n_1}{n_t} GI_{CA} + \frac{n_2}{n_t} GI_{CB} \right]$$

Can you make the process of finding the Best C for numerical feature fast ?

(1gm – 1000 gm)





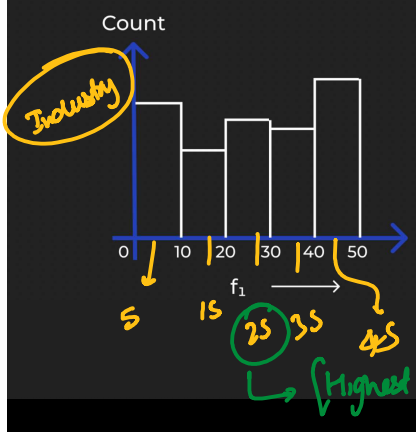
(1gm — 1000 gm)

140 crores

How to make splitting of numerical feature efficient?

- By binning the feature values.

Suppose, f_1 value range : [0 , 50]

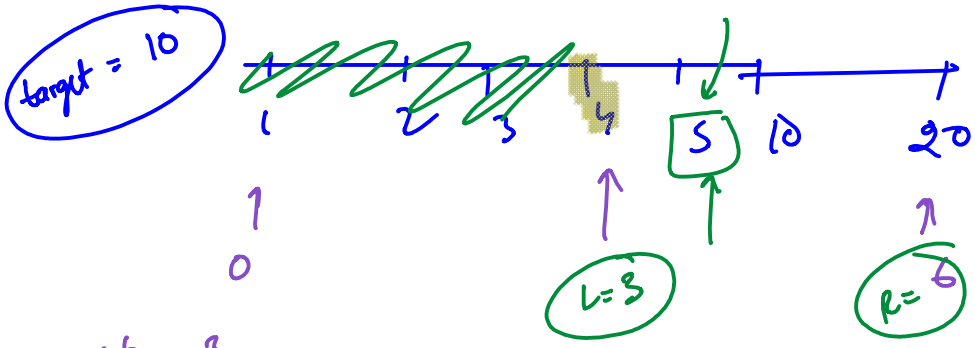


1. We create 5 bins for f_1 value and check IG value for the bins.
2. Select the bin for splitting which has maximum IG.

(Highest I.G gain)

20 — 30

25 → max I.G



$\frac{0+6}{2} = 3$

$\frac{3+6}{2} = \frac{9}{2} = [4]$

22:12

[overfit and Underfit]

[D-Tree] → never used in production

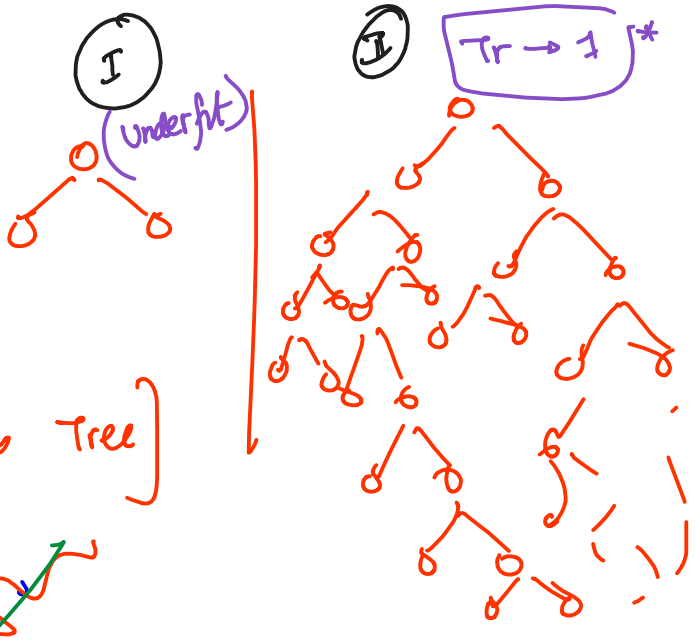
(1) → (overfit)

Train : 100%

Test : 70%

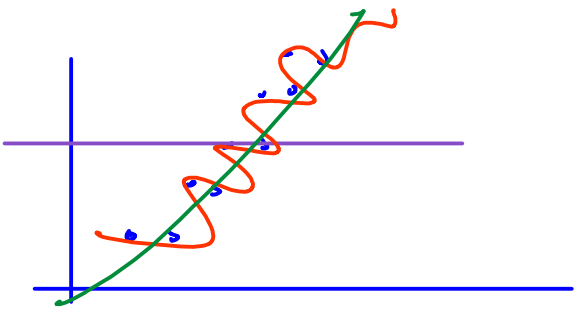
Tr → 100 (1)

Tes → 76%



Why D-Tree overfit?

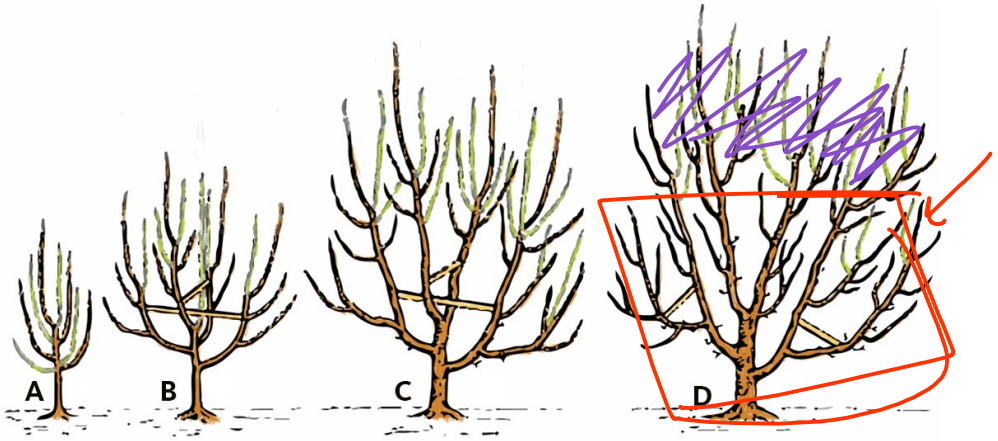
U/F
O/F
→ [Fn of depth of the Tree]



* [How to prevent tree from overfitting]

(#) [Limit the depth of the Tree]

Pruning →

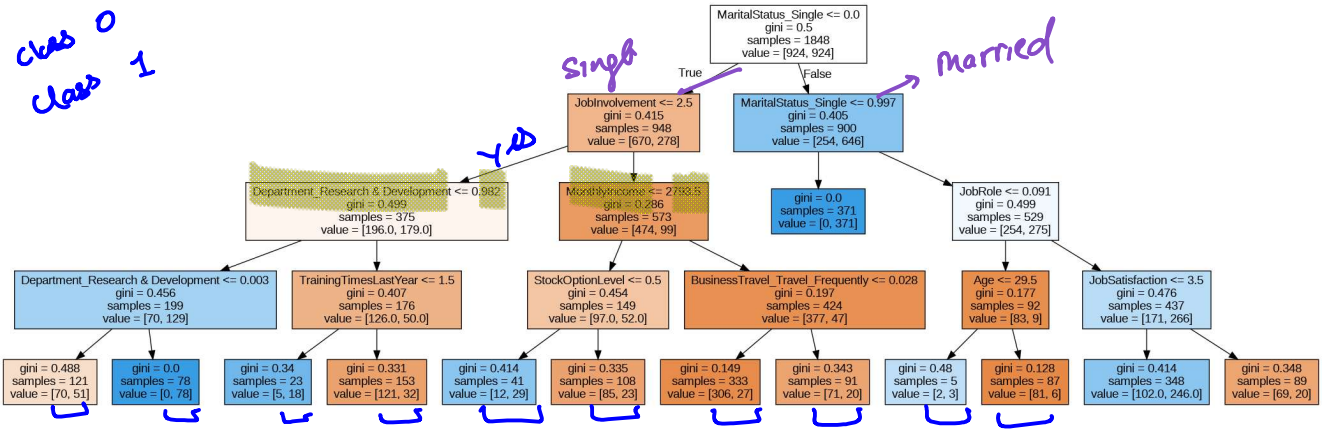


[Post Read H.W] → Read about Pruning

sklearn.tree.DecisionTreeClassifier

class sklearn.tree.**DecisionTreeClassifier**(*, *criterion='gini'*, *splitter='best'*, *max_depth=None*, *min_samples_split=2*, *min_samples_leaf=1*, *min_weight_fraction_leaf=0.0*, *max_features=None*, *random_state=None*, *max_leaf_nodes=None*, *min_impurity_decrease=0.0*, *class_weight=None*, *ccp_alpha=0.0*) [source]

* (Hyperparameter Tune D. Trees)



Practical Considerations of D. Trees (interview)

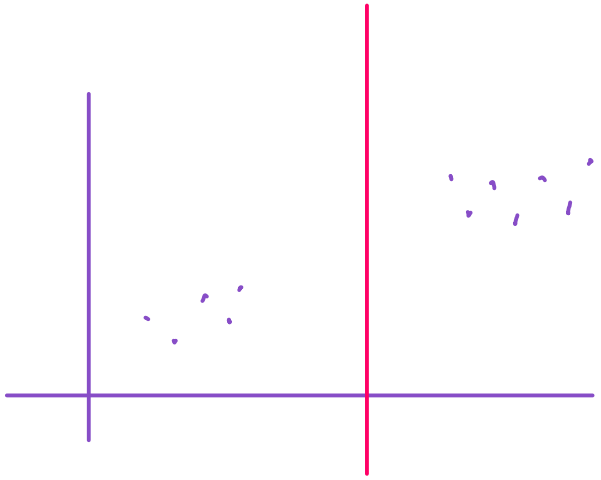
Which of the given DTs will be affected by an outlier?

4 options

Active Duration (Most preferred: 30 seconds)

Appears for: 45 Secs

A	DT with depth=50
B	DT with depth=5
C	Both of them
D	None of them



(1) Do we require Feature Scaling ?

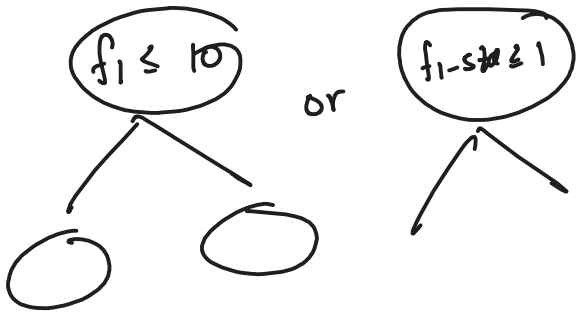
> Ans → NO

KNN
SVM
LR
NN

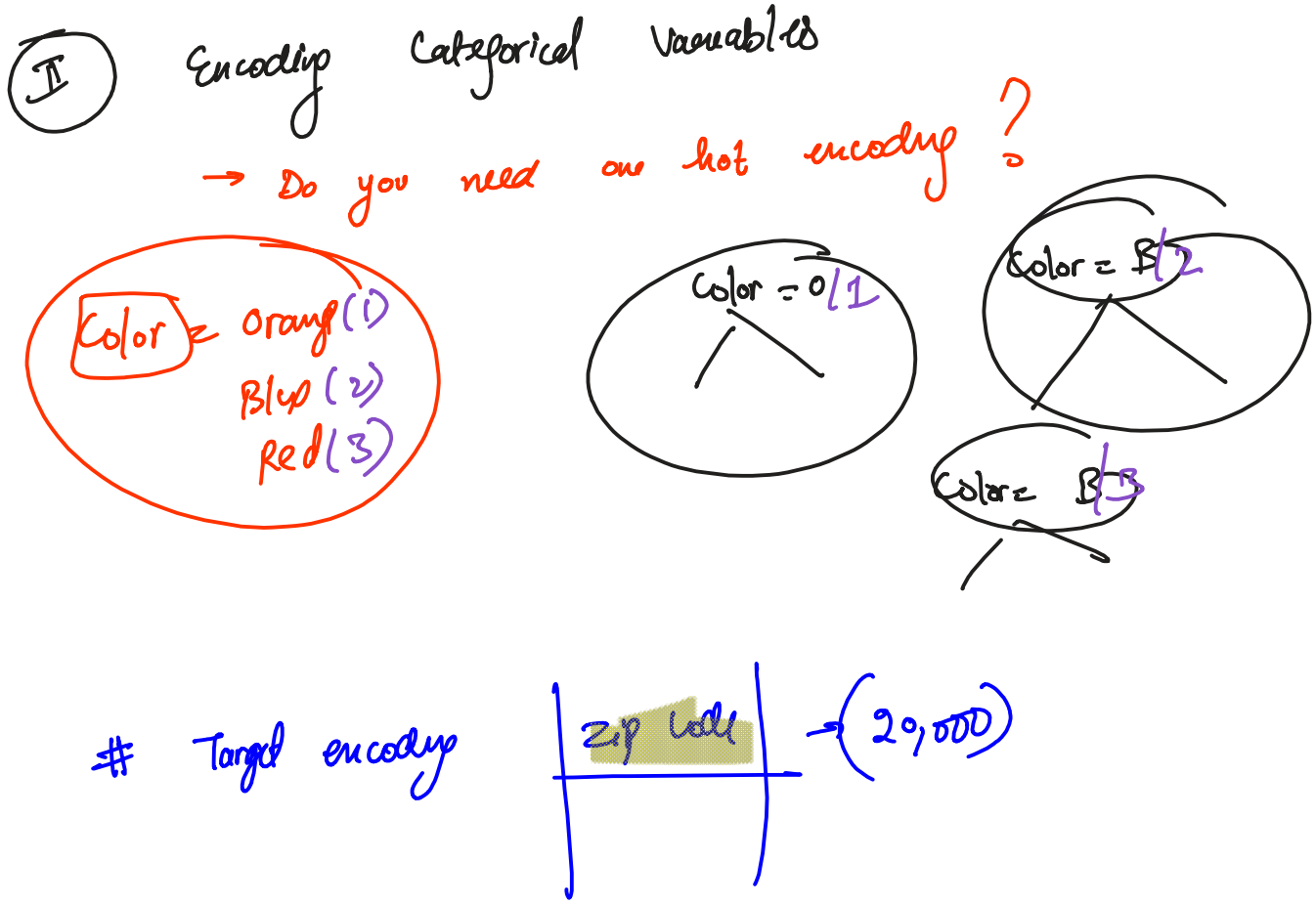
f
10
20
30

→

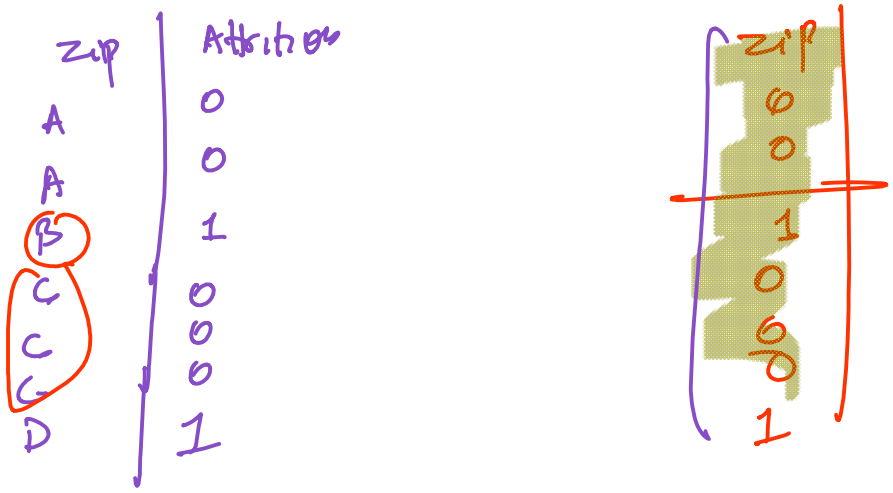
f_std
-1
-1.5
-2



But **Decision Trees** don't depend on the magnitude — they depend on **splits based on thresholds** that **separate classes**.



Target Encode 20,000 → less no of unique values



Dont how to scale, dont how to encode (much)

IIT

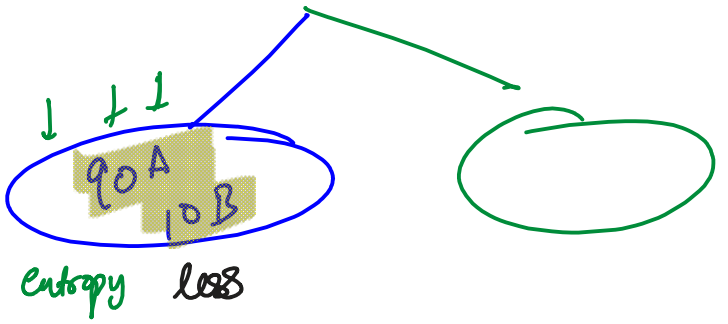
What about Imbalance data

90% → A

10% → B

good Results ?

No



we usually avoid imbalance data on Tree Based Algo

Fix

→ Balanced datasets (o/s, smote)

→ class weights (more imp to minority class)

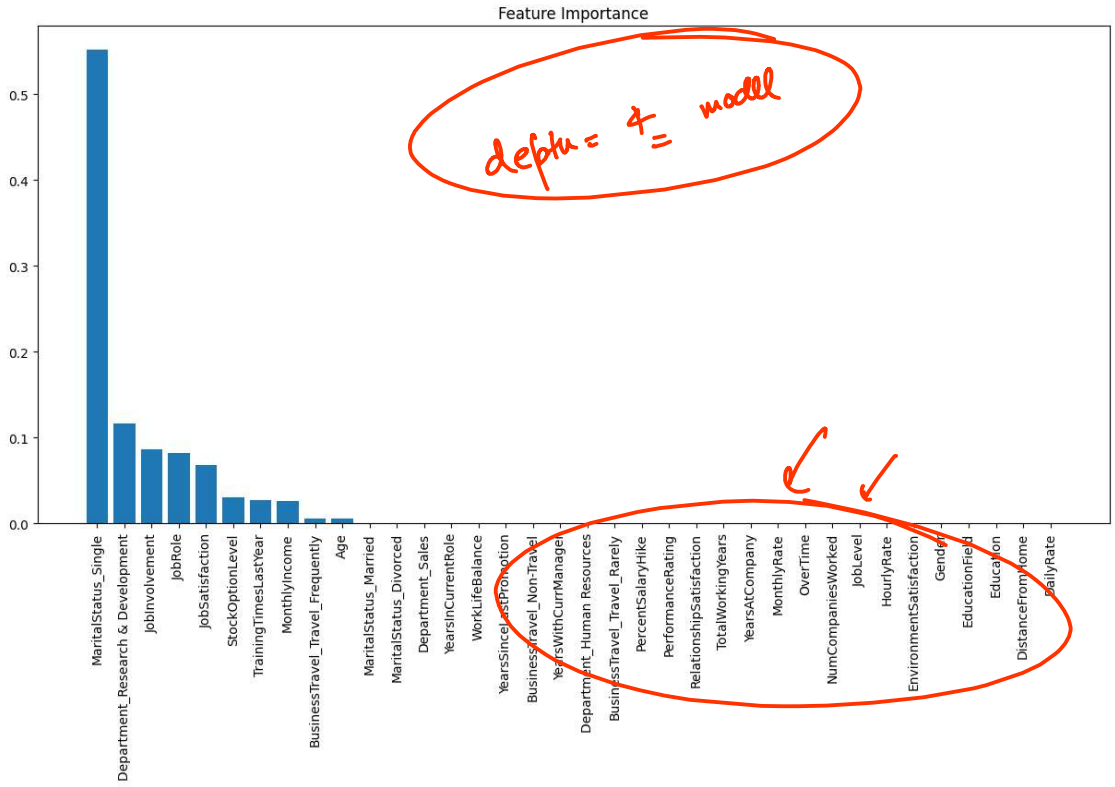
→ ~~Accuracy~~ P, F1, ROC-AUC

Feature Importances

Total Informet Gain for X_j (2 Splits) (normalised)

$$FI_{(X_j)} = IG_1 \frac{(n_1)}{n_T} + IG_{(2)} \frac{n_2}{n_T}$$

(2 spaces)
(Normalised)



x x

(*) (D. Tree for Regression) → (Pre_Read)

* ((High Dim) (X) → slow)

Assumptions
Tree Based
Models